

The aggressive strategy (A) corresponds to higher cost but low risk. The normal strategy (B) is associated with a mid-level of cost and a mid-level of risk. The passive strategy (C) is a low cost strategy but does have higher risk. Selection of a strategy such as D would not be appropriate because there are alternative strategies with the same risk but lower cost, lower risk at the same cost, or both lower cost and lower risk. A strategy such as D is deemed an irrational trading strategy and will never provide best execution.

Step 3—Specify Adaptation Tactic

The next step in the decision process consists of specifying how the algorithm is to adapt to changing market conditions. Algorithms may also include multiple types of adaptation tactics. Below are some common adaptation settings. Mathematical techniques behind these adaptation tactics are provided in Chapter 8.

Volume-based. Adjust the trading schedule based on market liquidity. A POV/participation rate algorithm is an example of a volume-based adaptation tactic. While these are often constant volume rates, they result in faster trading in times of higher market volume and slower trading in times of lower market volume.

Price-based. Adjust the trading schedule based on market prices. Aggressive-in-the-Money (AIM) algorithms that trade faster in times of favorable prices and slower in times of adverse price movement, and Passive-in-the-Money (PIM) algorithms that trade slower in times of favorable prices and faster in times of adverse price movement are types of price-based scaling algorithms.

Time-based. The algorithm adjusts its trading rate to ensure executions by a specified time, such as no later than the close. This algorithm may well finish sooner than specified but will not finish any later.

Probabilistic. Determines the appropriate trading rate to always provide the best chances (highest likelihood) of achieving the investment objective. It is based on a non-linear optimization technique, such as maximizing Sharpe ratio or minimizing tracking error.

Optimization Technique. The trade schedule is continuously adjusted so that its expected finishing price will be within a specified tolerance. These types of algorithms will often be based on a variation of a z-score measure (see Chapter 9) and incorporate realized costs (past), sunk cost or savings (dependent upon price increase or decrease since commencement of